Roadmap for climate-neutral ships - Emissions pathway

A.P.Møller - Maersk's
World's first carbon neutral liner in operation (Bio-methanol / VLSFO)

EU-funded ShipFC project
World's ammonia-powered fuel cell on offshore vessel with 2MW (Viking Energy, Equinor)

ESVAGT - Ørsted
World's first zero emission service operation vessel (SOV) operates on green fuels (e-methanol)

AMOGY - World's First Ammonia-Powered, Zero-Emission Ship
World's first zero emission ammonia-powered, zero-emission tugboat

Wallenius Wilhelmsen
World's first wind-powered RoRo (Orcelle Wind)

HyShip project
New roll-on/roll-off running on liquid green hydrogen (Topeka, Enova funding)

Ulstein – Nedstack - GE
Zero-Emission DP2 Construction Support Vessel Full LH_2-powered fuel cell with 2MW (Ulstein SX190)

A.P.Møller-Maersk's
Zero CO_2 emissions per container transported (today 46%)

Business-as-usual
(Based on predictions by IMO due to increasing sea traffic, 4% annual growth)

Technical improvements

Operational improvements

Wind propulsion

Battery

Other technologies / fuels

Hydrogen + Ammonia

Methanol

LNG / LPG

GHG reduction by

Alternative Technologies / fuels for replacing conventional fuels

Emissions pathway

- **Energy efficiency technology** (wind assistance, air lubrication, waste heat recovery, …)
- **Energy saving device** (wake equalizing duct, boss cap fin, …)
- **GHG-Emissions reduction devices** (Scrubbers, …)

- **Operational and environmental performance of the vessel** (fuel comsuption, speed reduction, transported cargo and distance travelled)

- **Plasma technology produces a synthetic natural gas (SNG)**
- **Anaerobic digestion of varying biomass sources to produce biomethane**
- **Carbon capture and storage on-board**

Scenario-based assessment of multiple source::
- Maritime forecast to 2050, Energy Transition Outlook 2020, DNV GL
- Assessment of selected alternative fuels and technologies , DNV GL June 2019
- Low carbon shipping towards 2050, DNV GL
- The maritime energy transition: Power-to-X as a key element for green shipping, MTU-Webinar, 12.11.2020
- Roadmap for Zero-Emission of International Shipping, MLIT Japan

2008

2020

2022

2025

2030

2050

LOHC-testing offshore
200kW unit on-board Edda Ferd (ENOVA funding)

EU ETS 2024 to 2026 *
CO_2 tax for maritime
40% → 70% → 100%
Including CH_4 & N_2O

IMO GHG
CO_2 Reduction 40% compared to 2008
EU 2030 target
40% → 55%
Reduction of CO_2 compared 1990
Norway 2030
CO_2 tax (x 2) for most industries

LOHC-Ready on-Board (SHIP-AH2OY project)
Østensjø 1st CSOV with Hybrid power propulsion system

MPEC 80 Revision
mid-2023
100% decarbonisation by 2050?

IMO GHG (2050)
GHG Reduction 50% compared to 2008
CO_2 Reduction 70% compared to 2008
EU 2050 target
Net zero

IMO's GHG strategy

Zero emissions
As soon as possible

International Maritime Organization (IMO)

Emissions

Time

Short term measures by IMO
EEDI, EEXI, SEEMP (CII, AER, EEOI)

Mid- and long term measures by IMO
Methane emission, EEDI (Phase 4) / Alternative fuels

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on-board